

Lecture #25,26

Content to be covered

- ❖ Wave function
 - ❑ Its physical interpretation
 - ❑ Its properties
- ❖ Operators
- ❖ Expectation values
- ❖ Some problems

Resources to be consulted

- ❑ Concepts of Modern Physics by Arthur Beiser

Matter Waves...Quantitative ; Wave Function

The wave associated with particles are mathematical constructs. It does not describe the space –time variation of any measurable quantity like displacement or any other characteristics present in the medium.

The wave relates to the probabilities $|\psi(x, y, z, t)|^2$ of observing the particle at different space locations as a function of time. In quantum mechanics, a wave function (ψ) describes wave characteristics of a particle (particularly, de-Broglie wave).

Physical interpretation of Wave function

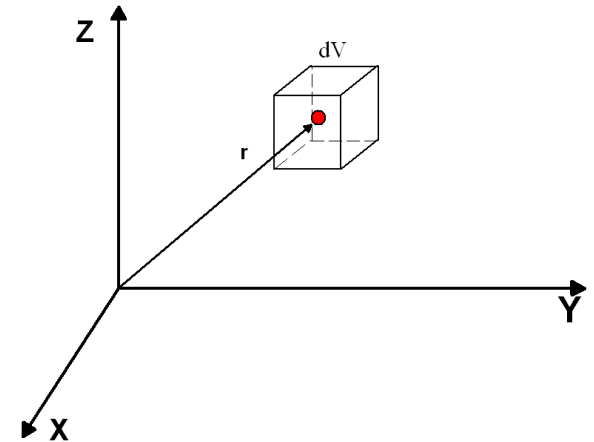
- The probability $P(\mathbf{r},t)dV$ to find a particle associated with the wavefunction $\Psi(\mathbf{r},t)$ within a small volume dV around a point in space with coordinate $\mathbf{r}$ at some instant t is

$$P(\vec{r},t)dV = |\psi(\vec{r},t)|^2 dV$$

– $P(\mathbf{r},t)$ is the probability density

- For one-dimensional case

$$P(x,t)dx = |\psi(x,t)|^2 dx$$



Wave function may be a **complex quantity**, but its mod square will always be a positive quantity. Hence,

$$|\psi(\vec{r},t)|^2 = \psi^*(\vec{r},t)\psi(\vec{r},t)$$

Probability density

$$\text{If } \psi = \psi_0 e^{-i(\omega t - \mathbf{k} \cdot \mathbf{r})} \text{ and } \psi^* = \psi_0 e^{-i(\mathbf{k} \cdot \mathbf{r} - \omega t)}$$

$$\text{now } \psi^* \psi = |\psi|^2 = \psi_0^2$$

Physical interpretation of Wave function

- The probability of finding a particle somewhere in a volume V of space is

$$\int_V P(\vec{r}, t) dV = \int_V |\psi(\vec{r}, t)|^2 dV$$

- Since the probability to find particle anywhere in space is 1, we have condition of normalization

$$\int_{allspace} |\Psi(r, t)|^2 dV = 1$$

normalized wave function

For **1-Dimensional case**, the probability of finding the particle in the arbitrary interval $a \leq x \leq b$ is

$$P_{ab} = \int_a^b |\psi(x, t)|^2 dx$$

$$P_{all\ space} = \int_{-\infty}^{+\infty} |\psi(x, t)|^2 dx$$

Properties of wave function

An acceptable/well behaved wave function Ψ must fulfill the following requirements:

- Ψ must be finite and single valued
- Ψ must be continuous and have a continuous first derivative everywhere.
- Ψ must be normalized *i.e.* it means that Ψ & Ψ^* must go to 0 at the boundaries (*i.e.* $x = \pm\infty$ or at the boundaries).

$$\int \psi^* \psi dV = \int |\psi|^2 dV = N \text{ (FINITE VALUE)}$$

If $N = 1$, wave function is NORMALIZED.

Once we know this wavefunction we know “everything” about the system!

Problems

Example 1

- A system is defined by the wave function:

$$\psi_{(x)} = A \cos\left(\frac{2\pi x}{L}\right) \text{ for } -\frac{L}{4} \leq x \leq \frac{L}{4}$$

- (a) What is the normalisation constant 'A'.
- (b) Find the probability between in the region $0 \leq x \leq \frac{L}{8}$

Sol.: (a)

$$\int_{-L/4}^{L/4} |\psi|^2 dx = A^2 \int_0^{L/8} \cos^2(2\pi x / L) dx = 1$$

$$\Rightarrow A^2 \cdot L / 4 = 1 \text{ or } A = 2 / \sqrt{L}$$

$$\text{(b) } p = \int_0^{L/8} |\psi|^2 dx = A^2 \int_0^{L/8} \cos^2(2\pi x / L) dx$$

$$\Rightarrow p_{(0-L/8)} = \int_0^{L/8} \frac{4}{L} \cos^2(2\pi x / L) dx$$

$$= \frac{4}{L} \left(\frac{(2 + \pi)L}{16\pi} \right) \approx 0.41$$

Example 2

A wave function is described as below:

$$\psi_{(x)} = A \sin(kx) + B \cos(kx) \text{ for } -\frac{L}{2} \leq x \leq \frac{L}{2}$$

- (a) Find the normalization constant and
- (b) Probability density

Sol: (a) Apply boundary conditions:

$$\psi_{(-L/2)} = 0 \Rightarrow -A \sin(kL/2) + B \cos(kL/2)$$

$$\psi_{(L/2)} = 0 \Rightarrow A \sin(kL/2) + B \cos(kL/2)$$

$$\Rightarrow 2B \cos(kL/2) = 0 = 2 \cos(2n+1)\pi/2$$

$$\Rightarrow k = \frac{(2n+1)\pi}{L}$$

The wave function:

$$\psi_{(n,x)} = A \sin((2n+1)\pi x / L)$$

Normalization will yield:

$$\int_{-L/2}^{L/2} |\psi|^2 dx = A^2 \int_{-L/2}^{L/2} \sin^2(2n+1)\pi x / L dx = 1$$

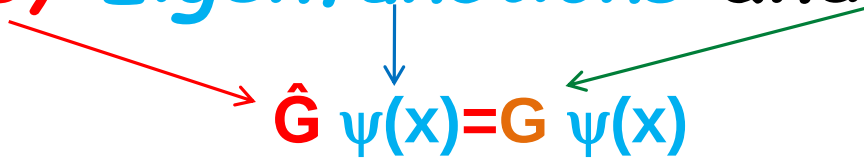
$$\Rightarrow A^2 = 2/L \text{ or, } A = \sqrt{2/L}$$

Probability density:

$$\psi^2_{(n,x)} = 2/L \sin^2((2n+1)\pi x / L)$$

Operators

Operators, Eigenfunctions and eigenvalues


$$\hat{G} \psi(x) = G \psi(x)$$

An operator is an instruction; a symbol which tells us to perform one or more mathematical operations on a function, say $f(x)$. The essential point is that they act on a function.

- ❖ Operators act on everything to their right, unless the action is constrained by brackets.
- ❖ Addition and subtraction rules : $(\hat{G}_1 \pm \hat{G}_2)f(x) = \hat{G}_1f(x) \pm \hat{G}_2f(x)$
- ❖ The product of two operators implies successive operation. $\hat{G}_1\hat{G}_2f(x) = \hat{G}_1[\hat{G}_2f(x)]$
- ❖ The product of two operator will result a third operator. $\hat{G}_1\hat{G}_2 = \hat{G}_3$
- ❖ Two operator commute if they follow $\hat{G}_1\hat{G}_2 = \hat{G}_2\hat{G}_1$

$$[\hat{G}_1, \hat{G}_2] = \hat{G}_1\hat{G}_2 - \hat{G}_2\hat{G}_1 = 0$$

Some Operators in Quantum Mechanics

**Physical Quantity(observable) → Operators
symbol ↔ actual operation**

In quantum mechanics, there exists an operator corresponding to each dynamical observable of classical physics. For examples:

Position	$x \rightarrow \hat{x} \leftrightarrow x$
Linear Momentum	$p_x \rightarrow \hat{p}_x \leftrightarrow -i\hbar \frac{\partial}{\partial x}$
Momentum in 3D	$p \rightarrow \hat{p} \leftrightarrow -i\hbar \nabla$
Total Energy	$E \rightarrow \hat{E} \leftrightarrow i\hbar \frac{\partial}{\partial t}$
Kinetic Energy	$K\hat{E} \rightarrow \frac{\hat{p}^2}{2m} \leftrightarrow -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2}$
Potential Energy	$V(x) \rightarrow \hat{V}(x) \leftrightarrow V(x)$
Total energy (Hamiltonian form)	$H \rightarrow \hat{H} \leftrightarrow \left[\frac{\hat{p}^2}{2m} + V \right]$

- These operators may be represented by matrices.
- These operators satisfy Eigen value equations of the form

Expectation Value

- Only **average value of physical quantities** can be determined (can't determine exact value of a quantity at a point)
- These average values are called Expectation Value.
 - ❖ These are values of physical quantities that quantum mechanics predicts and which, from experimental point of view, are averages of multiple measurements
- Example, [expected] position of the particle.

$$\langle \hat{G} \rangle = \frac{\int_{-\infty}^{+\infty} \Psi^*(x, t) \hat{G} \Psi(x, t) dx}{\int_{-\infty}^{+\infty} \Psi^*(x, t) \Psi(x, t) dx} = \int_{-\infty}^{+\infty} \Psi^*(x, t) \hat{G} \Psi(x, t) dx$$

Expectation Value for Momentum of a Free Particle

In general

$$\langle p \rangle = \frac{\int_{-\infty}^{+\infty} \psi^*(x) \left[-i\hbar \frac{\partial}{\partial x} \right] \psi(x) dx}{\int_{-\infty}^{+\infty} \psi^*(x) \psi(x) dx} = \frac{\int_{-\infty}^{+\infty} \psi^*(x) \left[-i\hbar \frac{\partial}{\partial x} \right] \psi(x) dx}{1}$$

$$\langle p \rangle = i\hbar \int_{-\infty}^{+\infty} \psi^*(x) \cdot \frac{\partial \psi(x)}{\partial x} dx$$

For a free particle
($\psi = Ae^{ikx}$)

$$\langle p \rangle = \frac{\int_{-\infty}^{+\infty} [Ae^{ikx}]^* \left[-i\hbar \frac{\partial}{\partial x} Ae^{ikx} \right] dx}{\int_{-\infty}^{+\infty} [Ae^{ikx}]^* [Ae^{ikx}] dx}$$

$$\langle p \rangle = \frac{\int_{-\infty}^{+\infty} [Ae^{ikx}]^* (-i\hbar) ik [Ae^{ikx}] dx}{\int_{-\infty}^{+\infty} [Ae^{ikx}]^* [Ae^{ikx}] dx} = \frac{\hbar k \int_{-\infty}^{+\infty} [Ae^{ikx}]^* [Ae^{ikx}] dx}{\int_{-\infty}^{+\infty} [Ae^{ikx}]^* [Ae^{ikx}] dx} = \hbar k = p$$

Problems

1. $\langle x \rangle$ for a particle in a box of length L where $\psi_n(x) = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}$ $0 \leq x \leq L$

$$\langle x \rangle = \frac{\int_0^L \psi_n(x)^* x \psi_n(x) dx}{\int_0^L \psi_n(x)^* \psi_n(x) dx} = \frac{2}{L} \int_0^L x \sin^2 \frac{n\pi x}{L} dx = \frac{2}{L} \frac{L^2}{4} = \frac{L}{2}$$

2. $\langle p \rangle$ for a particle in a box of length L . Use same the wave function as in the above problem.

$$\langle p \rangle = \frac{\int_0^L \psi_n(x)^* \hat{p} \psi_n(x) dx}{\int_0^L \psi_n(x)^* \psi_n(x) dx} = \frac{\int_0^L \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L} \left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \right) \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L} dx}{\int_0^L \frac{2}{L} \sin^2 \frac{n\pi x}{L} dx}$$

3. A Particle limited to the x -axis has the wave function $\psi = ax$ between $x=0$ and $x=1$; $\psi = 0$ elsewhere.

(a) Find the probability that the particle can be found $x=0.45$ and $x=0.55$.

(b) Find the expectation value $\langle x \rangle$ of the particle's position.

$$0.0251 a^2, a^2/4$$

Schrödinger's equation of Motion

In 1926, Schrödinger presented his famous wave equation as development of de-Broglie ideas of the wave properties of matter. The Schrödinger's equation is the fundamental equation of quantum mechanics in the same sense as the Newton's second law of motion of classical mechanics. It is the differential equation for the de Broglie waves associated with particles and describes the motion of particles.

The wave function is ψ for a particle moving freely in the x-direction is given as

$$\psi = Ae^{-i\omega(t-x/v)}$$

In terms of energy E and momentum p :

Since $v=E/h$ (Planck's relation) and $\lambda=h/p$ (de-Broglie relation)

$$\psi(x, t) = A e^{-(2\pi i/h)(Et-px)}$$

Schrödinger's equation of Motion

- Schrödinger developed the wave equation which can be solved to find the wavefunction by translating the equation for energy of classical physics into the language of waves

$$\frac{p^2}{2m} + V = E \quad \longleftrightarrow \quad -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V\psi = i\hbar \frac{\partial \psi}{\partial t}$$

- For fixed energy, we obtain the **time-independent** Schrödinger equation, which describes stationary states

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V\psi = E\psi$$

- the energy of such states does not change with time
 - $\psi_n(x)$ is an eigen function or eigen state
 - V is a potential function representing the particle interaction with the environment

Schrödinger Wave Equation: Time Independent

Position and time dependent wave function can be described as below:

$$\psi(r, t) = \psi_0 e^{-i(\omega t - k \cdot r)} = [\psi_0 e^{-i\omega t}] e^{ik \cdot r} = [\psi_0 e^{ik \cdot r}] e^{-i\omega t}$$
$$\nabla^2 \psi = \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} = -\frac{\omega^2}{v^2} \psi \quad \frac{\omega^2}{v^2} = \left[\frac{2\pi\nu}{\nu\lambda} \right]^2 = \frac{4\pi^2 p^2}{h^2} = \frac{2m(E - V)}{\hbar^2}$$
$$\nabla^2 \psi = -\frac{\omega^2}{v^2} \psi = -\frac{4\pi^2 p^2}{h^2} \psi \quad \text{as } p^2 = 2m(E - V)$$
$$\text{or } \nabla^2 \psi = -\frac{4\pi^2}{h^2} [2m(E - V)] \psi$$
$$\Rightarrow \boxed{\nabla^2 \psi + \frac{2m}{\hbar^2} (E - V) \psi = 0}$$

The above equation is known as time independent Schrödinger wave equation.

for a free particle $V=0$

$$\boxed{\nabla^2 \psi + \frac{2mE}{\hbar^2} \psi = 0}$$

$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2m}{h^2}(E - V)\psi = 0$$

Where $\Psi(x,t)$ is called wave function. The differential equation of matter wave in one dimension is derived as above.

The above equation is called one-dimensional Schrödinger's wave equation in one dimension. In three dimension, the Schrödinger wave equation becomes

$$\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2} + \frac{8\pi^2m}{h^2}(E - V)\psi = 0$$

$$\text{or } \nabla^2\psi + \frac{8\pi^2m}{h^2}(E - V)\psi = 0$$

Schrödinger Wave Equation: Time Dependent

Time independent Schrodinger wave equation can be illustrated as below:

$$\nabla^2 \psi + \frac{2m}{\hbar^2} (E - V) \psi = 0$$

$$\text{or } -\nabla^2 \psi + \frac{2mV}{\hbar^2} \psi = \frac{2mE}{\hbar^2} \psi$$

$$\Rightarrow -\frac{\hbar^2}{2m} \nabla^2 \psi + V\psi = E\psi$$

as we know Energy ($\hat{E}$) $\rightarrow i\hbar \frac{d}{dt}$

$$\text{or } E\psi = i\hbar \frac{d\psi}{dt}$$

Substituting the value of Energy operator, we may get:

$$-\frac{\hbar^2}{2m} \nabla^2 \psi + V\psi = i\hbar \frac{\partial \psi}{\partial t}$$

$$\Rightarrow \left[\frac{\hbar^2}{2m} \nabla^2 + V \right] \psi = i\hbar \frac{\partial \psi}{\partial t}$$

The above equation is known as Time dependent Schrödinger equation.

Here, $\hat{H} \equiv \left[\frac{\hbar^2}{2m} \nabla^2 + V \right]$ known as Hamiltonian Operator

Hence,

$$\left[\frac{\hbar^2}{2m} \nabla^2 + V \right] \psi = E\psi \Rightarrow \hat{H}\psi = E\psi$$